

2023



Southwest Airlines Scales App Development Using AWS Device Farm & Amazon EC2 Mac Instances

Learn how Southwest Airlines accelerated development of new mobile app features using AWS Device Farm and Amazon EC2 Mac Instances.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#)

4x

increase in build speed by using Amazon EC2 Mac Instances

Migrated

its entire iOS build fleet to Amazon EC

Removed

the need to manage numerous physical devices



Hi, I can connect you with an AWS representative or answer questions you have on AWS.



Need more info? Highlight any text to get an explanation generated with AWS generative AI.



7.1x

acceleration of test-run speed for



Saves

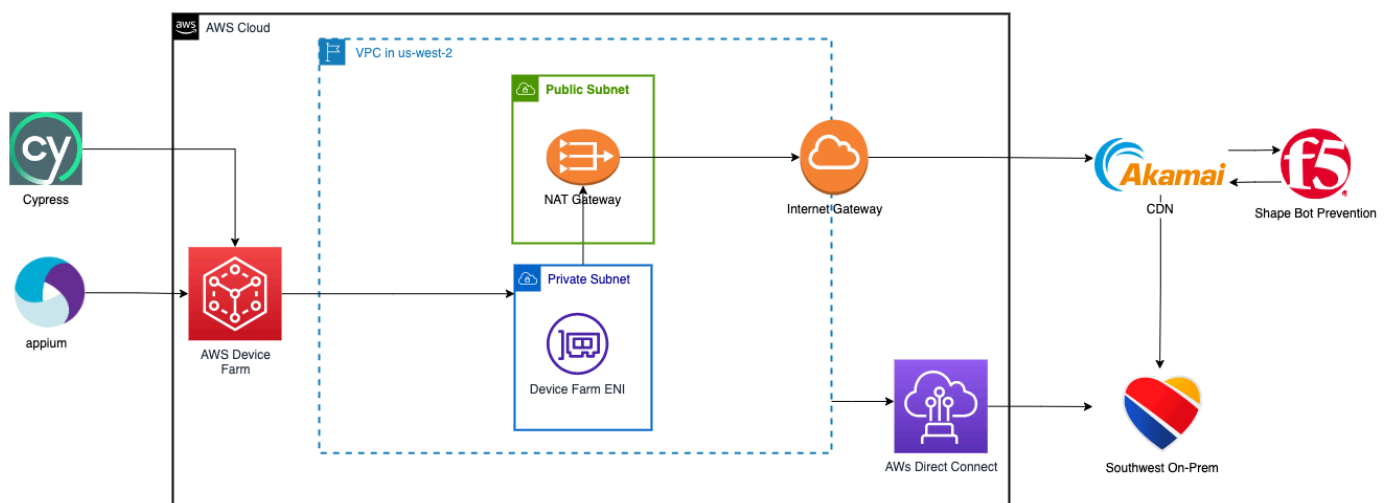
cost by finding and fixing errors earlier in the development lifecycle

Overview

Prioritizing efficiency and customer satisfaction, [Southwest Airlines](#) (Southwest) has become one of the world's largest low-cost airlines. Key to meeting those goals is allowing fliers to self-serve on Southwest.com and the mobile app for Apple and Android devices.

Providing high performance on those solutions is business critical. Southwest tests every update, new feature, and change before deploying it to customers. It also checks that changes to the app do not degrade functionality or performance.

To scale application development, Southwest migrated its continuous integration and continuous delivery (CI/CD) pipeline, including Mac computers and mobile devices, from on-premises infrastructure to Amazon Web Services (AWS) using GitLab. Now, developers can test applications and improve user experience on virtually any device or operating system that customers might use.



Opportunity | Using AWS Device Farm to Automate the App Testing Process for Southwest

Southwest flies 130 million passengers a year to 120 airports in 11 countries. Facilitating a great travel experience for such a high number of customers is challenging. That is why Southwest focuses on efficient operations and self-service options. Today, its fliers can use both the desktop and mobile applications to book, change, and cancel flight, car, and hotel reservations and upgrade boarding. Soon, they will be able to track their bags and receive more real-time travel updates. Customers are increasingly using self-service.

Expanding the functionality of Southwest's applications means likewise scaling their release process. Previously, app testing had to be manually performed on a physical device; the company used numerous small devices, and

Mac computers to test every application change. Faulty devices often resulted in testing delays caused by waiting for new ones. Moreover, the COVID-19 pandemic made swapping and procuring devices more difficult.

Southwest needed a solution that would facilitate automated, test-driven development and avoid the handoff of devices between testing teams. It chose [AWS Device Farm](#), an application testing service that organizations use to improve their web and mobile apps by testing them across a range of desktop browsers and real mobile devices without having to provision and manage a testing infrastructure. AWS Device Farm maintains physical devices on behalf of Southwest, automatically confirming that each device is functional and ready for tests on demand. “AWS Device Farm is very user-friendly and easily accessible,” says Sundeep Bobba, cloud and DevOps tech lead on Southwest.com digital platforms.

Southwest began using AWS Device Farm in 2019 and uses it today for test automation across various platforms. From AWS Device Farm, Southwest can use public and private devices with ease. These devices are exclusively for Southwest’s use and provide connectivity to the company’s on-premises environment. AWS Device Farm also generates reports, including screenshot videos that testers can save as documentation.

Although the Southwest team is now using AWS Device Farm devices for manual testing, it is also writing application-performance-monitoring pipelines on GitLab to automate testing on the devices. Using GitLab and AWS Device Farm, teams can share test functions, centralize results, and derive analytics on how to improve their overall testing function as much as possible.

Southwest also paired AWS Device Farm with Appium, an open-source automation tool for running functional tests on devices. That accelerated test-run speed by an average of 7.1 times for new features: 2.3 times for iOS, 2.5 times for Android, 6 times for iPadOS, and 15 times for browsers. The company uses AWS Device Farm to simply scale its automation test workloads of Appium tests to dozens of device types and operating systems and versions of those systems.

Solution | Increasing Build Speed by 4x Using Amazon EC2 Mac Instances for Southwest

Southwest did away with its entire fleet of Mac computers by 2021 and now relies on [Amazon Cloud](#) (Amazon EC2) [Mac Instances](#), which organizations use to run on-demand macOS workloads in the cloud.



“There was huge applause when we achieved that, because we can now run and manage builds on demand,” says Bobba. Now, the Southwest team can automatically provision Amazon EC2 Mac Instances in minutes.

After migrating the Apple iOS CI/CD pipeline to Amazon EC2 Mac Instances, build speed increased by four times. Using the more scalable, flexible infrastructure on AWS, Southwest supported the same development lifecycle for its mobile and desktop apps. That gives customers the same experience on a browser as on the mobile app.

On AWS, Southwest can accelerate the delivery of new features to customers. “Using Amazon EC2 Mac Instances and AWS Device Farm, we find and fix issues earlier in the pipeline,” says Mona Pinjani, senior manager of technology at Southwest. “That saves cost because fixing errors later in production is expensive.”

Southwest also improved security by using Amazon EC2 Mac Instances and [AWS Secrets Manager](#), which organizations can use to manage, retrieve, and rotate database credentials, API keys, and other secrets throughout their lifecycles. “We consume secrets on the fly when we are running the builds, then we delete them,” says Bobba.

Outcome | Meeting the Goal of Efficiency on AWS

Now that Southwest has a modernized, streamlined CI/CD pipeline on AWS, it plans to unbundle app releases and instead perform targeted deployments. “The goal is to use AWS Device Farm and Amazon EC2 Mac Instances in our CI/CD pipeline to transition into on-demand releases,” says Pinjani. Southwest also plans to use those AWS services in browser-based testing.

“On AWS, we provide the infrastructure, DevOps tools, and resources that a developer needs, meeting our goal of being better, faster, and more efficient,” says Pinjani.

About Southwest Airlines

Southwest Airlines was founded in 1967 and has become one of the world’s largest low-cost airlines, with 75,000 employees managing flights to 120 airports in 11 countries for 130 million passengers per year.

AWS Services Used

AWS Device Farm

AWS Device Farm is an application testing service that lets you improve the quality of your web and mobile apps by testing them across an extensive range of desktop browsers and real mobile devices; without having to provision and manage any testing infrastructure.

[Learn more »](#)



Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 750 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

[Learn more »](#)

Amazon EC2 Mac Instances

Amazon Elastic Compute Cloud (Amazon EC2) Mac instances allow you to run on-demand macOS workloads in the cloud for the first time, extending the flexibility, scalability, and cost benefits of AWS to all Apple developers.

[Learn more »](#)

AWS Secrets Manager

AWS Secrets Manager helps you manage, retrieve, and rotate database credentials, API keys, and other secrets throughout their lifecycles.

[Learn more »](#)

More Travel and Hospitality Customer Stories

AUSTRALIA



Lonely Planet cut itinerary generation costs by 80%...

Lonely Planet, is a premier travel media company tested multiple generative AI solution vendors and found Amazon

CANADA



Using Amazon AppStream 2.0 to Increase Scalability...



Bedrock to be 78% more cost-effective for their travel use case. Further to that, they were able to generate experiences and itineraries at 80% less the cost of manually curating them.

2024

Learn how Air Canada in the travel industry gained scalability and reliability while optimizing costs by migrating its on-premises VDI to AWS.

2024

IRELAND



Ryanair on AWS

Supporting better employee and customer experiences, running the business with greater efficiency, and preparing the runway for growth.

2024

AMERICAS



United Airlines on AWS

United Airlines is leading modernization in an industry that's embracing change. The largest airline globally by number of seat miles, its mission is to "Connect People, Uniting the World." Migrating tens of hundreds of applications to Amazon Web Services (AWS) has helped the business accelerate innovation, even in the midst of challenging market conditions. From supporting its technology team to save \$2 billion and boosting developer productivity 40%



2024

Get Started

Organizations of all sizes across all industries are transforming their businesses and delivering on their missions every day using AWS. Contact our experts and start your own AWS journey today.



Learn About AWS

- What Is AWS?
- What Is Cloud Computing?
- AWS Accessibility
- AWS Inclusion, Diversity & Equity
- What Is DevOps?
- What Is a Container?
- What Is a Data Lake?
- What is Artificial Intelligence (AI)?
- What is Generative AI?
- What is Machine Learning (ML)?
- AWS Cloud Security
- What's New
- Blogs
- Press Releases

Resources for AWS

- Getting Started
- Training and Certification
- AWS Solutions Library
- Architecture Center
- Product and Technical FAQs
- Analyst Reports
- AWS Partners

Developers on AWS

- Developer Center
- SDKs & Tools
- .NET on AWS
- Python on AWS
- Java on AWS
- PHP on AWS
- JavaScript on AWS

Help

- Contact Us
- Get Expert Help
- File a Support Ticket
- AWS re:Post
- Knowledge Center
- AWS Support Overview
- Legal
- AWS Careers



Amazon is an Equal Opportunity Employer: *Minority / Women / Disability / Veteran / Gender Identity / Sexual Orientation / Age.*

Language

- عربي |
- Bahasa Indonesia |
- Deutsch |
- English |
- Español |
- Français |
- Italiano |
- Português |



Tiếng Việt |
Türkçe |
Русский |
ไทย |
日本語 |
한국어 |
中文 (简体) |
中文 (繁體)

Privacy
|
Accessibility
|
Site Terms
|
Cookie Preferences
|

© 2024, Amazon Web Services, Inc. or its affiliates. All rights reserved.

